Transfer task: plan and begin

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Organize the given evidence and begin an independent position prediction.

SUCCESS CRITERIA - I can...

1. Organize the task data with fixed axes and matching units.
2. Calculate central and bounded final coordinates.
3. Save an explicit plan for the graph and explanation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “bound” to explain your model or result. Sentence starters are optional.

component: the part of a vector along one chosen axis

bound: an upper or lower limit in a model

assumption: a condition accepted for using a model

01 Warm-up: bring back an earlier idea

A 3 kg lamp hangs at rest from a vertical rope.
Use g=9.8.
Find tension. Then find tension if it accelerates upward at 2 m/s².

Word help: tension = the pulling force through a stretched rope or cable

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

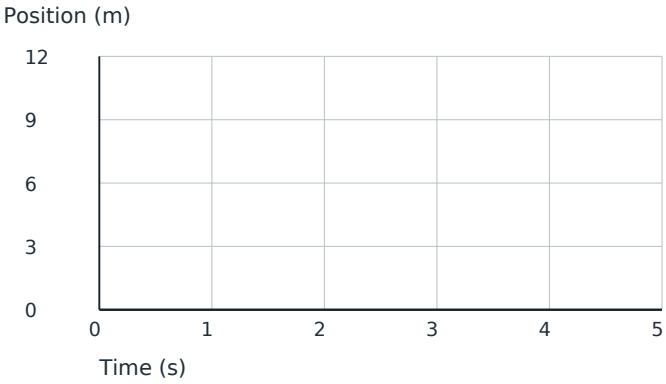
02 Worked example: explain the model

Reference: $v_x = v \cos\theta$; $v_y = v \sin\theta$; $x_f = x_i + v_x t$
$y_f = y_i + v_y t$.
Fictional fixed-Earth model: $x_i = -1.2\times10^8$ km; $y_i = 0$; $v = 8500$ m/s; $\theta = 15°$ from +x toward +y; $t = 1.6\times10^7$ s. Convert units.

Worked reasoning: A simplified model uses x_i=0 m and constant v=2 m/s for 5 s, giving x=10 m. This is a conditional prediction, not proof that the real object follows the model.

Your turn: Draw the predicted path and mark one observation you would collect to test the constant-velocity assumption.

Sentence starter: I predict ... . My model supports this because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Plan and represent

A fictional asteroid model fixes Earth at the origin and ignores gravity and Earth's motion.
Use the givens to plan the unit conversion, velocity components, final position, bounds and graph.
Sketch fixed +x and +y axes, with +x initially toward Earth.

Sentence starter: My design shows … . It meets the requirement by … .

Quantity	Value
Initial x	-1.2×10^8 km
Initial y	0 km
Speed	8500 m/s
Direction	15° from +x toward +y
Elapsed time	1.6×10^7 s

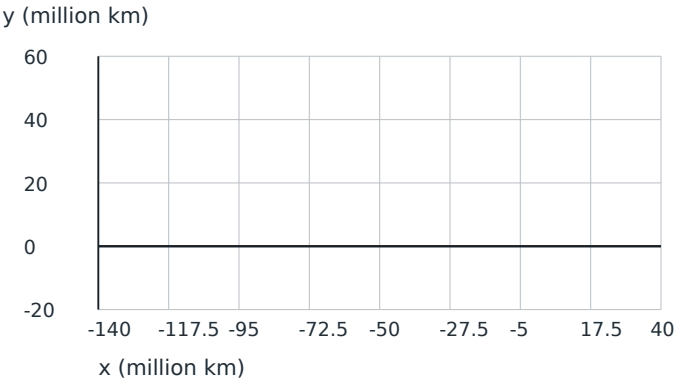
Marzano check - circle: ① ② ③

04 Central prediction

Resolve the velocity in question 3 and calculate final x and y in kilometers. Keep unrounded components until the end. Plot the path and distinguish final position, displacement and distance from the origin.

Word help: resolve = separate a vector into components

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Propagate bounded inputs

The initial x bound is ±50,000 km and the speed bound is ±200 m/s at the same 15° angle. Initial x and speed vary independently; angle, initial y and time are exact.
Find final coordinate half-widths and intervals.

Sentence starter: I found … [unit]. This value means … in this situation.

Coordinate	Half-width (km)	Lower (km)	Upper (km)
x	____	____	____
y	____	____	____

Marzano check - circle: ① ② ③

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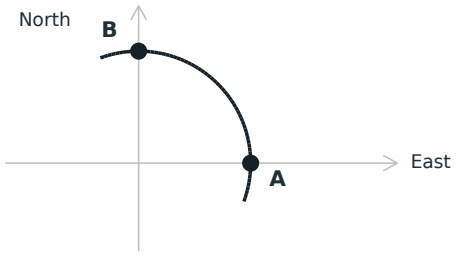
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A puck follows a circle at a steady 7 m/s. A student calls it equilibrium.
Draw velocity at two points and correct the claim.
Use the curve for the turning case; show how velocity changes.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Path sketch: runner moves from A toward B.

Marzano check - circle: 1 2 3

07 Checkpoint

Prepare the central, lower and upper x values at the start and final time for tomorrow's graph.
Use questions 3-5.
State one assumption, one omitted effect, and the next step you need to finish.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Time	Central x	Lower x	Upper x
Start	____	____	____
Finish	____	____	____

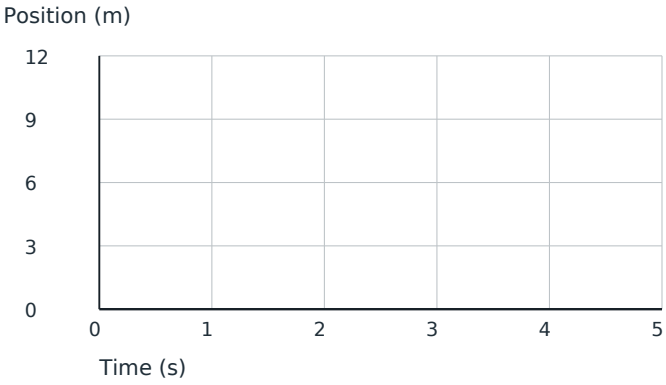
Marzano check - circle: 1 2 3

08 STRETCH: test the model

A model predicts x=10 m while the measurement is 8.0±0.2 m. Construct one constant-velocity model and one uniformly accelerated model from rest that fit a 5 s trip to 8 m.
Explain what more data would resolve.

Word help: resolve = separate a vector into components

Sentence starter: I predict ... . My model supports this because ... .



Marzano check - circle: 1 2 3